In bad news for Democritus and his friends, nineteenth-century physics didn't explain the world in terms of particles alone. It suggested, instead, that two fundamental kinds of stuff were required: both particles and fields.

Fields can be thought of as the opposite of particles, at least in the context of classical mechanics. The defining feature of a particle is that it's located at one point in space, and nowhere else. The defining feature of a field is that it is located everywhere. A field is something that has a value at literally every point in space. Particles need to interact with each other somehow, and they do so through the influence of fields.

Think of the magnetic field. It's a *vector field*—at every point in space it looks like a little arrow, with a magnitude (the field can be strong, or weak, or even exactly zero) and also a direction (it points along some particular axis). We can measure the direction in which the magnetic field points just by pulling out a magnetic compass and observing what direction the needle points in. (It will point roughly north, if you are located at most places on Earth and not standing too close to another magnet.) The important thing is that the magnetic field exists invisibly everywhere throughout space, even when we're not observing it. That's what fields do.

There is also the electric field, which is also a vector with a magnitude and a direction at every point in space. Just as we can detect a magnetic field with a compass, we can detect the electric field by placing an electron at rest and seeing if it accelerates. The faster the acceleration, the stronger the electric field.* One of the triumphs of nineteenth-century physics was when James Clerk Maxwell unified electricity and magnetism, showing that both of these fields could be thought of as different manifestations of a single underlying "electromagnetic" field.

The other field that was well known in the nineteenth century is the gravitational field. Gravity, Isaac Newton taught us, stretches over astronomical distances. Planets in the solar system feel a gravitational pull toward the sun, proportional to the sun's mass and inversely proportional to the square of the distance between them. In 1783 Pierre-Simon Laplace showed that we can think of Newtonian gravity as arising from a "gravitational potential field" that has a value at every point in space, just as the electric and magnetic fields do.